Rational Number Project

Fraction Operations and Initial Decimal Ideas **Lesson 9.5: Overview**	**Materials**
Students name models for decimals on a 10 x 10 grid using decimal notation.	• 10 x10 grid with 3 orange strips and 4 yellow squares shaded and one with 34 yellow squares shaded • Student Pages A, B • Orange and yellow colored crayons, pencils or markers for each student

Teaching Actions

Warm Up

Using a two 10 x 10 grids represent 47-hundredths in two different ways. Name this amount in at least two ways.

Large Group Introduction

1. Show students a 10 x 10 grid that has 3 orange strips shaded plus 4 yellow squares. Ask: What part of the grid is covered? What are the different ways we can describe that amount:
   - $\frac{3}{10}$ and $\frac{4}{100}$
   - 3-tenths and 4 hundredths more

   Show the amount using 34 yellow squares, and ask: how can you describe this amount?
   - 34 out of 100
   - $\frac{34}{100}$
   - 34-hundredths

2. Explain: You can also record that amount covered in this way: 0.34

3. Ask: How can you make sense of this way of naming that amount by comparing 0.34 to this way of writing the amount? $\frac{3}{10} + \frac{4}{100}$. (Connect this to

Comments

As you work through these decimal lessons consider these 5 indicators of what it means for students to understand initial decimal ideas.

(1) use accurate language for decimal numbers, (2) use models with understanding to represent decimal numbers, (3) order decimals by composing and decomposing decimals using mental images of the models, (4) use their understanding of the relative size of decimals to guide their estimation with operations with decimals, and (5) interpret addition and subtraction operations with a model and using the model to build meaning for work with symbols.

The decimal lessons develop these understandings. Please note that using decimal language is important. Avoid the "point 2, 3" language for 23-hundredths. Students who can use accurate decimal language and be flexible on naming decimals avoid typical errors students make when comparing decimals or operating with decimals.

As you monitor students' learning, ask questions related to the 5 indicators. Some students will need more time than others to build mental

Teaching Actions

showing 3 orange strips and 4 yellow squares).

4. Ask: How can you make sense of this way of naming that amount by comparing .34 to 34 out of 100 or $\frac{34}{100}$? (Connect this to showing the amount as 34 yellow squares).

12. Possible explanations:
    - 3 is in the tenths place so it is $\frac{3}{10}$; 4 is in the 100ths place so it is $\frac{4}{100}$
    - If you consider how many 100ths are covered then it's 34-hundredths.

13. Repeat for the other examples on from lesson 9, Student Page B. Students should record the different ways discussed directly on their papers:
    - Name verbally the amount on the grid
    - Describe verbally to their partner what they see
    - Record as fractions in two ways
    - Write as a decimal

Small Group/Partner Work

14. Students complete Student Pages A and B. They are to represent each number given in two ways. They are to record as a decimal and in at least two other ways.

Wrap Up

15. As students complete the student pages, monitor students to see if they are using accurate decimal

Comments

images for decimals; they may struggle to not use whole number language for decimals; they may misinterpret the model and focus only on color (.5 is 5 orange squares while .06 is 6 yellow squares).

During class discussions look for students who do use language appropriately and understand the decimal model to share their thinking. Be purposeful in selecting students so examples of the 5 indicators are made public for all students to consider.

As you work through naming decimals remember that you want students to compose and decompose decimals. For example students should see .34 as the sum of two parts:
$\frac{3}{10}$ and $\frac{4}{100}$.

You want students to see the number as a single entity as well: $\frac{34}{100}$.

.34 = $\frac{3}{10} + \frac{4}{100}$

.34 = 34 out of 100 or $\frac{34}{100}$

Teaching Actions	Comments
language and if they can explain the meaning for the decimal by connecting to each 10 x 10 grid. 16. Select students purposefully to present how they drew two representations for each number and explain what the decimal symbol means by connecting back to the 10 x 10 grids	

Translations:

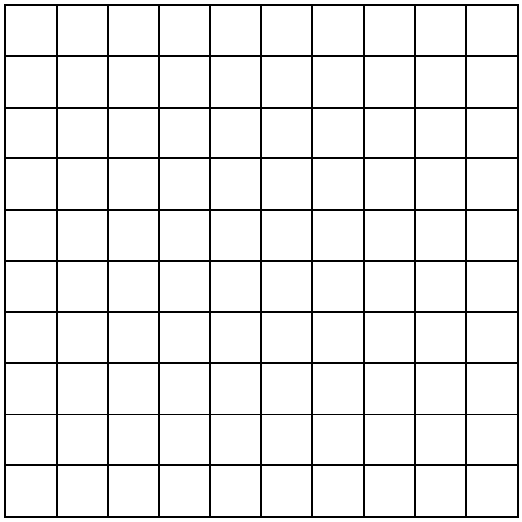
- Concrete to verbal
- Concrete to verbal to symbols
- Concrete to verbal to pictures to symbols
- Symbols to pictures to symbols

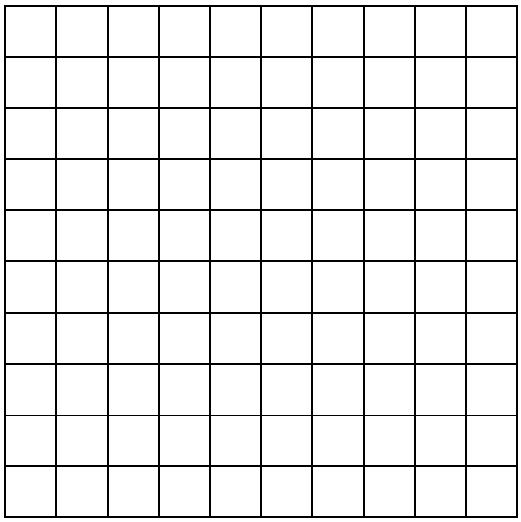
Using a two 10 x 10 grids represent 47-hundredths in two different ways. Name this amount in at least two ways.

Naming Tenths and Hundredths on a 10 x 10 Grid

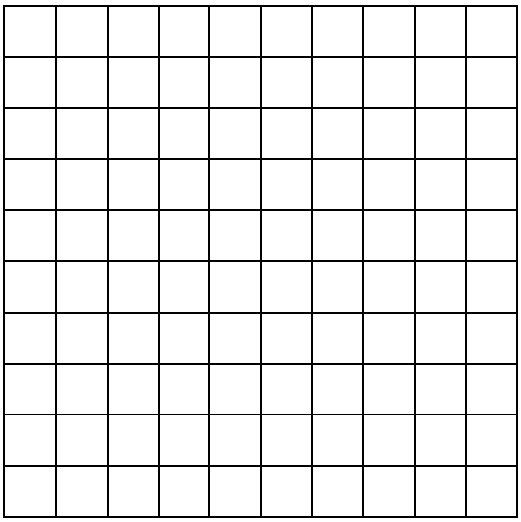
Show each number in two different ways on the 10 x 10 Grid. Then name the amount as a decimal and in two other ways of your choice.

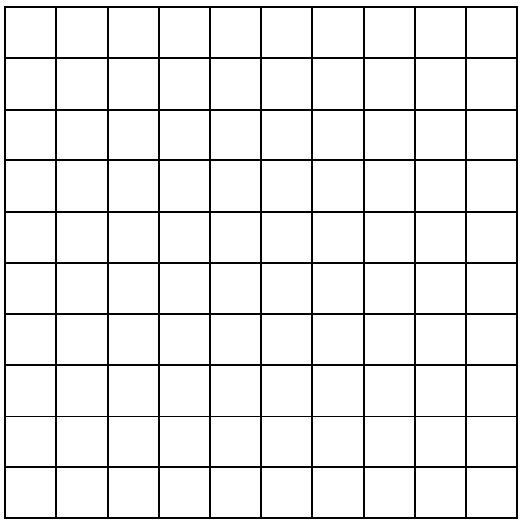
47-hundredths



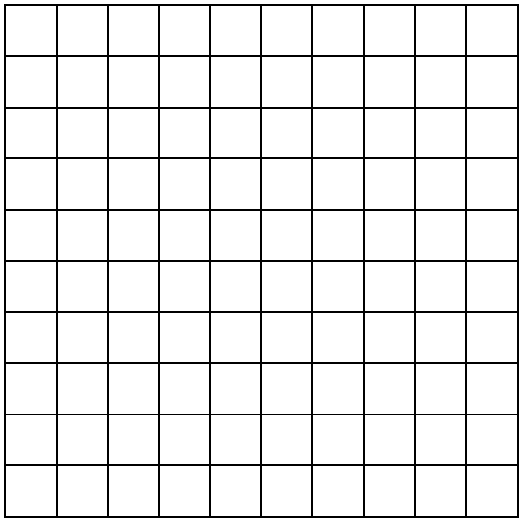


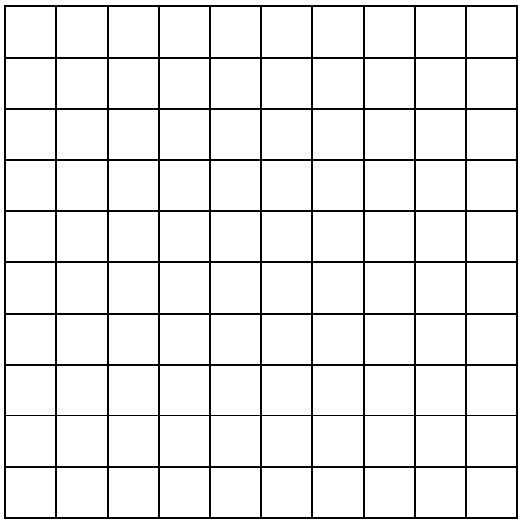
$\frac{4}{10}+\frac{3}{100}$



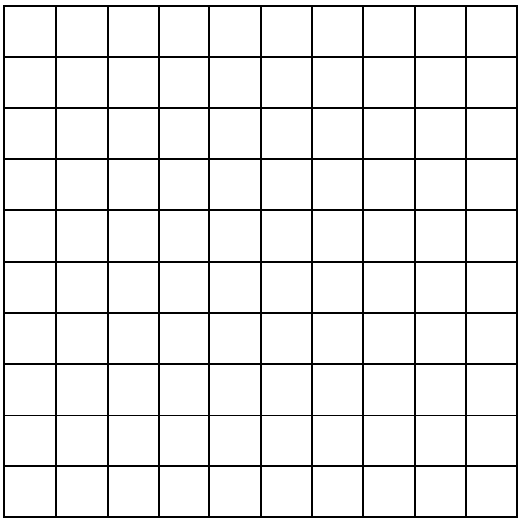


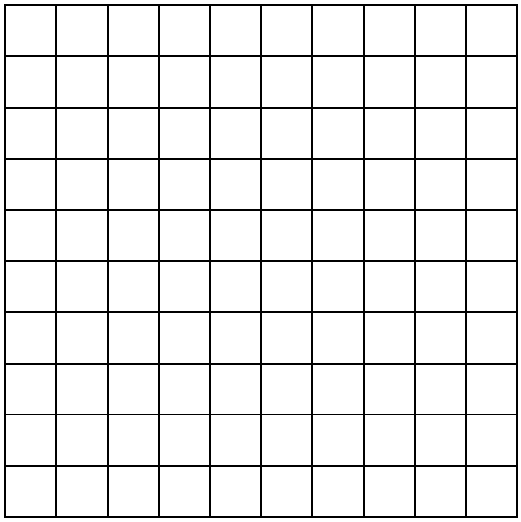
9-tenths and 8 hundredths





$\frac{5}{10}$





Post Lesson Reflection

Lesson_____

1) Number of class periods allocated to this lesson: _____

2) Student Pages used: _____

3) Adaptations made to lesson: (For example: added extra examples, eliminated certain problems, changed fractions used)

4) Adaptations made on Student Pages:

5) To improve the lesson I suggest: